

MICHIGAN DEPARTMENT OF ENVIRONMENT, GREAT LAKES, AND ENERGY

INTEROFFICE COMMUNICATION

TO: Claire Lowe, Cadillac District Office, Water Resources Division

FROM: Glen Schmitt, Permits Section, Water Resources Division

DATE: April 19, 2022

SUBJECT: Local Limits Review

MiWaters Submission Number: HPG-XCC7-NCG09

Cadillac Wastewater Treatment Plant (WWTP)

National Pollutant Discharge Elimination System (NPDES) Permit No. MI0020257

Pursuant to your request, we have reviewed the Industrial Pretreatment Program Local Limits request dated April 13, 2022 for the Cadillac WWTP. The facility is currently authorized to continuously discharge treated municipal wastewater from Monitoring Point 001A through Outfall 001 to the Clam River at Latitude 44.2657, Longitude -85.3985. A design flow of 3.2 million gallons per day (MGD) was used in determining the final effluent limitations and monitoring requirements in the current NPDES permit, but this flow is not considered a limitation or actual capacity.

The Clam River has a lowest 95 percent exceedance low flow, harmonic mean low flow, and 90dQ10 low flow of 2.6 cubic feet per second (cfs), 13 cfs, and 3.4 cfs (Low Flow File Number 10194). An ambient total hardness value for the Clam River of 70 milligrams per liter (mg/l) was used for the parameters that are total hardness dependent and was collected by Permits Section staff on August 19, 2021 from the Clam River at Latitude 44.263376, Longitude -85.4011.

The theoretical water quality-based effluent limits (WQBELs) are presented in micrograms per liter (ug/l) and pounds per day (lbs./day) for the parameters requested.

Parameter	CAS Number	Daily Maximum WQBEL (µg/L)	Daily Maximum Load (lbs./day)	Monthly Average WQBEL (µg/l)	Monthly Average Load (lbs./day)
Arsenic #	7440382	680	18	16	0.42
Cadmium	7440439	12	0.32	4.1	0.11
Chromium	7440473	1,300	34	94	2.5
Copper	7440508	29	0.77	11	0.29
Cyanide	57125	44	1.2	5.9	0.16
Lead	7439921	820	22	94	2.5
Mercury @	7439976	---	---	0.0013	3.5 x 10 ⁻⁵
Molybdenum	7439987	58,000	1,500	3,600	97
Nickel	7440020	760	20	47	1.3
Selenium	7782492	120	3.2	5.7	0.15
Silver	7440224	22	0.59	1.4	0.036
Zinc	7440666	360	9.7	210	5.5
Phenol	108952	6,800	180	510	14

Parameter	CAS Number	Daily Maximum WQBEL (µg/L)	Daily Maximum Load (lbs./day)	Monthly Average WQBEL (µg/l)	Monthly Average Load (lbs./day)
1,2-Dichlorobenzene	95501	240	6.4	15	0.39
Chloroform #	67663	11,000	290	710	19
Ethylbenzene #	100414	370	9.9	24	0.63
Lindane # @	58899	---	---	0.026	0.00069
Polychlorinated Biphenyls (PCBs) # @	1336363	---	---	0.000026	6.9 x 10 ⁻⁷
Styrene #	100425	2,900	77	130	3.5
Toluene	108883	2,600	69	290	7.8
Xylene	1330207	890	24	55	1.5

- Carcinogen

@ - bioaccumulative chemical of concern (BCC)

- 1) The current NPDES permit includes a monthly average total phosphorus limit of 0.5 milligrams per liter (mg/l) and a corresponding load limitation of 13 pounds per day (lbs./day) from May through March and a monthly average total phosphorus limit of 0.65 mg/l and a corresponding load limit of 17 lbs./day during the month of April. These limits are to protect the Clam River and downstream surface waters from excessive nutrients.
- 2) The recommendations contained herein are based on the Part 4 Water Quality Standards and Part 8 Water Quality-Based Effluent Limit Development for Toxic Substances. Treatment methods or economic concerns have not been addressed. We understand that other considerations may be used in deciding permit limits and requirements.

Please contact me at (517) 290-6424 or schmittg1@michigan.gov if you have any questions regarding the recommendations above.